3.10 Swivel toolhead

The toolhead of a shaper holds the cutting tool rigidly. It is fitted on the face of the ram. The vertical slide of the toolhead can be moved vertically or at a particular angle to provide vertical and angular feed movement to the tool. It allows the tool to have an automatic relief during the return stroke of the ram.

The toolhead has a swivel base attached to the circular seat on the ram. The swivel base has angular graduations marked on it. As the vertical slide is mounted on the swivel base of the toolhead, it may be set and moved at any desired angle to machine angular surfaces like 'V' grooves and dove tail grooves.

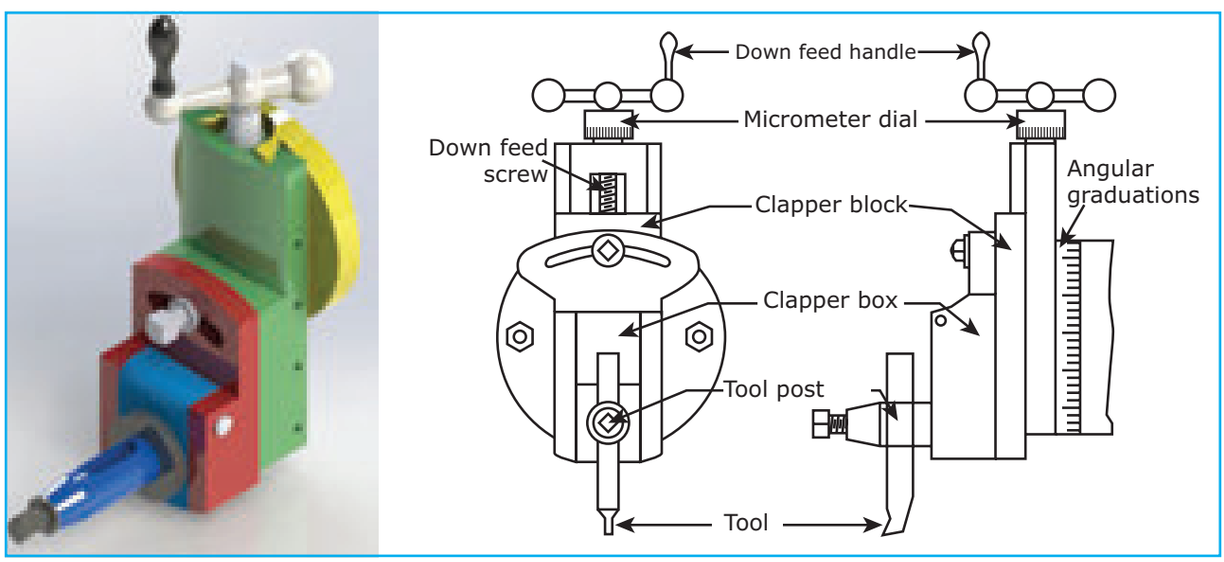
The downfeed screw handle is rotated to move the vertical slide up and down. A graduated dial is placed on the top of down feed screw to control the amount of depth of cut or feed accurately.

Apron consisting of clapper box, clapper block and tool post is clamped on the vertical slide by a screw. By releasing the clamping screw, the apron can be swiveled either towards left or towards right with respect to the vertical slide. The clapper box has two vertical walls within which the clapper block is housed. It is connected to the clapper box with the help of a hinge pin. This arrangement provides relief to the tool while machining vertical or angular surfaces. The tool post is mounted upon the clapper block. The tool post is provided with a slot to accommodate the tool and a screw to hold the tool rigidly on the tool post.

The clapper block fits securely inside the clapper box to provide a rigid tool support during forward stroke. On the return stroke, a slight frictional drag of the tool on the work lifts the block out of the clapper box and prevents the tool cutting edge from dragging on the work surface. Fig. illustrates the swivel toolhead of a shaper.

3.11 Work Holding Devices

Workpieces can be held and supported on the shaper table directly or by having some special devices. Depending on the size and shape of the work, it may be



Swivel Tool Head

XII BME_English version CHAPTER 03.indd 66 1/8/2020 1:43:57 PM